Higgs field itself. The strength of that interaction determines the particle's mass. The photon continues to zip through space at the speed of light, but the particles with mass move more slowly in proportion to the tugging they experience from the Higgs.

Comparing the behavior of particles in the early universe to their behavior today is like comparing how you might interact with vapor versus liquid water. Imagine the vapor is the Higgs field—an energy field, present at every point in space. And imagine that at some point that Higgs field drastically changes character, as completely as vapor condensing into liquid water. If you've been used to encountering nothing but humid air, moving through a pool of water is a completely different prospect. When the Higgs field suddenly shifted in character, it was as though the laws of physics had condensed into a totally different form. Suddenly, particles that could move through space unimpeded at the speed of light were slowed down by their interactions with the Higgs field. They obtained mass.

We call this process *electroweak symmetry breaking*.

FEARFUL SYMMETRY

Symmetry in physics is the kind of subtle, abstract concept that is extremely hard to explain without equations but which is also so absolutely vital to everything we think about as physicists that I can't in good conscience just brush by it. Symmetry is central to how we describe theories of nature, and, more often than not, to how we develop new ones. If you happen to be someone who is used to thinking about the world in terms of the mathematical equations that govern it, you're probably already comfortable with the idea that theories can be described in terms of the symmetries they obey; if you're not, that's complete gibberish to you, and understandably so. So let's take a short detour for a moment and lay this all out, because it's an incredibly beautiful thing, and once you know about it you see it everywhere.

Symmetry isn't just about whether or not something looks the same in a mirror. In physics, it's all about patterns, and how those patterns can give you deeper insight into some underlying structure. Take, for instance, the periodic table of the elements. Why are the elements arranged in the rows and columns we are used to seeing them in today? If you've studied chemistry, you'll know that certain columns collect elements that have